Yihong Zhu

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Education

Stanford University Jun 2025 – Present
Ph.D., Aeronautics & Astronautics | Advisor: Prof. Charbel Farhat **GPA: 4.0/4.0**
Research: computational mechanics (fluids/structures), projection-based model reduction, physics-based surrogate modeling

University of California, San Diego Sep 2021 – Jun 2025
B.S., Aerospace Engineering, Minor in Mathematics **GPA: 3.967/4.000**

Skills

Numerical Methods: Reduced-Order Modeling, CFD (Computational Fluid Dynamics), Finite Volume, Finite Difference, Finite Element, Dynamic Substructuring, Operator Inference
Programming & HPC: C++, Python (NumPy/SciPy, scikit-learn), MATLAB, Slurm, Git, Linux, HPC
Engineering Software: Ansys Fluent, Ansys FEA, SolidWorks CAD/FEA, OpenRocket

Research Experience

Farhat Research Group, Stanford University | *Advisor: Prof. Charbel Farhat* Jun 2025 – Present
- Implemented a boundary-condition-enforcing feature in AERO-F, the group's parallel C++ CFD/ROM code, mitigating "snapping," a catastrophic failure mode of projection-based reduced-order models (ROMs).
- Developed a Python ROM codebase for dynamic substructuring, accelerated with randomized numerical linear algebra.

Computational Modeling and Flow Physics Lab, UCSD | *Advisor: Prof. Oliver Schmidt* Aug 2024 – Jun 2025
- Developed Grassmannian interpolation with frequency alignment for spectral POD bases to predict modes at new parameters.

Flame-Fluids-Energy Lab, UCSD | *Advisor: Prof. Abhishek Saha* Jun 2023 – Aug 2024
- Studied thermoacoustic instability in multi-flame Rijke tubes; applied PCA in MATLAB to derive instability warning signals.

Numerical Integrators Group, UCSD | *Advisor: Prof. Harsh Sharma* Sep 2022 – Aug 2023
- Derived a symplectic, time-adaptive variational integrator for nonconservative orbits (e.g., low-Earth orbit with drag).

Engineering Projects

CFD Solver Development | Stanford AA214 & UCSD MAE185 Mar 2024 & Mar 2026
- Built a 2D unstructured finite volume Euler solver in MATLAB (Roe flux, 2nd-order reconstruction, RK2 time integration) for supersonic ramp and shock tube flows.
- Implemented a 2D finite-difference compressible Navier–Stokes solver in MATLAB for Mach 4 flow over a flat plate.

Learned Nonlinear Constraint Modes for Substructuring | Stanford CS229 Jun 2026
- Learned nonlinear corrections to linear constraint modes for a parametric nonlinear heat equation, comparing quadratic operator inference against a neural network (MLP) closure on SVD-compressed inputs; trained on 1,200 2D finite-difference snapshots.
- MLP closure remained $< 10\%$ error up to mild nonlinearity, about $10\times$ and $100\times$ lower than quadratic and linear models.

C++/CUDA Finite-Difference Heat Solver | Personal Project Nov 2026
- Implemented and verified a 2D explicit heat-equation solver in C++/CUDA; benchmarked from 10K to 9M grid points.
- Reached ∼165 GB/s on an NVIDIA A30 MIG slice (∼70% of estimated peak); performance up to 9× a single CPU core.

Rocket Propulsion Laboratory, UCSD | *Propulsion Engineer; Lead Engineer, 2023–2025* Oct 2021 – Apr 2025
- Designed airfoil fins with Ansys Fluent CFD to reduce drag while maintaining stability through ascent.
- Built a MATLAB quasi-1D nozzle design tool and modeled combustion thermochemistry to predict engine performance.

Selected Publications

- C. Farhat, **Y. Zhu**, F. As'ad, T. Taddei. The Snapping Phenomenon and Boundary Enforcement in Model Reduction: Mechanisms and Mitigation. *Under review, CMAME.*
- **Y. Zhu**, S. Sato, O. Schmidt. SPOD Mode Interpolation with Frequency Alignment on the Grassmannian Manifold. *AIAA SciTech Forum, 2026.*
- Y. Weng, **Y. Zhu**, A. Saha. Vortex-driven Subharmonic Bifurcation in a Multi-flame Rijke Tube. *Combustion and Flame, 282, 114447, 2025.*
- **Y. Zhu**, H. Sharma. Hamiltonian Adaptive Variational Integrators for Nonconservative Problems in Astrodynamics. *AIAA Region VI Student Conference, 2024.*